

Program – 4: Integrate Git with Docker for Source-Controlled Application Builds

Description

This experiment demonstrates how to create a Node.js application, containerize it using Docker, push the source code to GitHub, and enable source-controlled application builds.

Prerequisites

- Git installed
- Docker installed
- Node.js and npm installed
- GitHub account

Project Structure

```
Program-4/  
|— Dockerfile  
|— index.js  
|— package.json  
|— package-lock.json
```

Procedure

Step 1: Create Project Folder

```
mkdir program-4  
cd program-4
```

Step 2: Initialize Node.js Project

```
npm init -y
```

Step 3: Create index.js

```
const express = require("express");  
const app = express();  
const PORT = 3000;  
app.get("/", (req, res) => {  
  res.send("Hello from Docker + Git!");  
});  
app.listen(PORT, () => {  
  console.log(`Server running at http://localhost:${PORT}`);  
});
```

Step 4: Install Dependency

```
npm install express
```

Step 5: Create Dockerfile

```
FROM node:18-alpine
```

```
WORKDIR /app
```

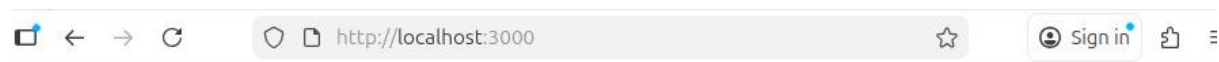
```
COPY package*.json ./
```

```
RUN npm install
```

```
COPY . .
```

```
EXPOSE 3000
```

```
CMD ["node", "index.js"]
```



Hello from Git & Docker Integrated!

Step 6: Create GitHub Repository

Create a new repository on GitHub (no README).

Step 7: Initialize Git Locally

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git branch -M main
```

Step 8: Link Remote Repository

```
git remote add origin https://github.com/<username>/<repository>.git
```

Step 9: Push to GitHub

git push -u origin main

The screenshot shows a GitHub repository page for 'manasa-s7 / Pro4'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows an initial commit by 'manasa-s7' 2 months ago, with files: node_modules, Dockerfile, index.js, package-lock.json, and package.json. The README section is empty, with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, and 0 forks. It also includes sections for Releases, Packages, Languages (JavaScript 57.1%, Dockerfile 42.9%), and Suggested workflows (Datadog Synthetics).

Pin sites you visit the most
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Pin this tab

manasa-s7 / Pro4

main 1 Branch 0 Tags

Go to file Add file Code

manasa-s7	initial commit	d9c153 · 2 months ago	1 Commit
node_modules	initial commit	2 months ago	
Dockerfile	initial commit	2 months ago	
index.js	initial commit	2 months ago	
package-lock.json	initial commit	2 months ago	
package.json	initial commit	2 months ago	

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

JavaScript 57.1% Dockerfile 42.9%

Suggested workflows

Based on your tech stack.

Datadog Synthetics Configure

Run Datadog Synthetic tests within your GitHub Actions workflow